

Gaussian random scalars

- Univariate normal distributions

Univariate normal

A Normal random variable is denoted with

$$w \sim \text{Normal}(\mu, v)$$

This variable attains real values in $(-\infty, +\infty) \equiv \mathbb{R}$

The probability density function is given by

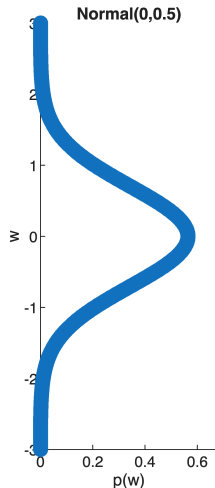
$$p(w) = \text{Normal}(w; \mu, v) = \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{1}{2} \frac{(w - \mu)^2}{v}\right)$$

The parameters are reals $\mu \in (-\infty, +\infty) \equiv \mathbb{R}$ and $v \in (0, \infty)$

- It **matters** because it is the basis for everything BO
- It is **simulated** by

$$w = \mu + \sigma * \text{randn};$$

which reads $w = \mu + \sigma\zeta$ and $\zeta \sim \text{Normal}(0, 1)$



Simulation

To generate a normal variate

$$w \sim \text{Normal}(\mu, v)$$

- First, **find** a σ such that

$$\sigma^2 = v$$

- Then, **generate** a standard normal variate

$$\zeta \sim \text{Normal}(0, 1)$$

- Finally, **transform**

$$w = \mu + \sigma\zeta$$

In this algorithm, we have **two choices**

$$\sigma = +\sqrt{v}$$

$$\sigma = -\sqrt{v}$$

$$w = \mu + \sqrt{v}\zeta$$

$$w = \mu - \sqrt{v}\zeta$$

The algorithm works because of the **de-whitening property**

$$\left. \begin{array}{l} \zeta \sim \text{Normal}(0, 1) \\ w = \mu + \sigma\zeta \end{array} \right\} \implies w \sim \text{Normal}(\mu, \sigma^2)$$

Transformation properties

De-whitening property

$$\left. \begin{array}{l} \zeta \sim \text{Normal}(0, 1) \\ w = \mu + \sigma \zeta \end{array} \right\} \implies w \sim \text{Normal}(\mu, \sigma^2)$$

Whitening property

$$\left. \begin{array}{l} x \sim \text{Normal}(\mu, v) \\ y = \frac{x - \mu}{\pm \sqrt{v}} \end{array} \right\} \implies y \sim \text{Normal}(0, 1)$$

General transformation property

$$\left. \begin{array}{l} x \sim \text{Normal}(m, v) \\ y = \alpha + \beta x \end{array} \right\} \implies y \sim \text{Normal}(\alpha + \beta m, \beta^2 v)$$

Other properties

Addition property

$$\left. \begin{array}{l} x_1 \sim \text{Normal}(m_1, v_1) \\ x_2 \sim \text{Normal}(m_2, v_2) \\ y = x_1 + x_2 \end{array} \right\} \implies y \sim \text{Normal}(m_1 + m_2, v_1 + v_2)$$

Magnitude properties

$$\left. \begin{array}{l} x_n \sim \text{Normal}(0, 1) \\ y = \sum_{n=1}^N x_n^2 \end{array} \right\} \implies y \sim \chi_N^2$$

$$\left. \begin{array}{l} x_n \sim \text{Normal}(0, 1) \\ z = \sqrt{\sum_{n=1}^N x_n^2} \end{array} \right\} \implies z \sim \chi_N$$

Important functions

The **cumulative distribution function** of $w \sim \text{Normal}(\mu, v)$ is

$$\begin{aligned} C(w) &= \int_{-\infty}^w dw' p(w') \\ &= \int_{-\infty}^w dw' \text{Normal}(w'; \mu, v) \\ &= \frac{1}{2} \left(1 + \text{erf} \left(\frac{1}{\sqrt{2}} \frac{w - \mu}{\sqrt{v}} \right) \right) \\ &= \Phi \left(\frac{w - \mu}{\sqrt{v}} \right) \end{aligned}$$

Our $C(w)$ is a **continuous** and **strictly increasing** function

$$C : \mathbb{R} \mapsto (0, 1)$$

The **quantile function** is the **inverse**

$$Q : (0, 1) \mapsto \mathbb{R}$$

$$\begin{aligned} Q(p) &= C^{-1}(p) \\ &= \mu + \sqrt{2v} \text{erf}^{-1}(2p - 1) \\ &= \mu + \sqrt{v} \Phi^{-1}(p) \end{aligned}$$

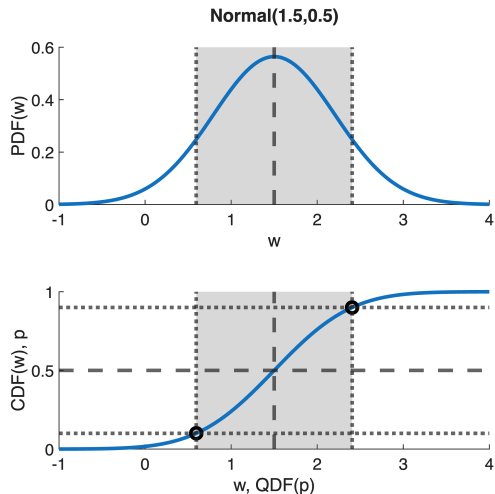
Credible intervals

A **credible set** is a set of values with a designated total probability of containing the value of our uncertain quantity

The γ -credible interval is a **symmetric interval** with a total probability $\gamma \in [0, 1]$

This is given by

$$\left[Q\left(\frac{1}{2} - \frac{\gamma}{2}\right), Q\left(\frac{1}{2} + \frac{\gamma}{2}\right) \right] \subset \mathbb{R}$$



Bayesian considerations

In Bayesian models, normals are more conveniently parametrized by **precision** $\tau = 1/\nu$

$$w \sim \text{Normal}(\mu, 1/\tau)$$

Unknown **mean**, known variance

$$\mu \sim \text{Normal}(M, 1/T)$$

$$w|\mu \sim \text{Normal}(\mu, 1/\tau)$$

The posterior is

$$\mu|w \sim \text{Normal}(M', 1/T')$$

$$M' = \frac{TM + \tau\mu}{T + \tau}$$

$$T' = \frac{1}{T + \tau}$$

Known mean, unknown **variance**

$$\tau \sim \text{Gamma}(\Phi, \Psi)$$

$$w|\tau \sim \text{Normal}(\mu, 1/\tau)$$

The posterior is

$$\tau|w \sim \text{Gamma}(\Phi', \Psi')$$

$$\Phi' = \Phi + \frac{1}{2}$$

$$\Psi' = \frac{1}{\frac{1}{\Psi} + \frac{(w-\mu)^2}{2}}$$

Unknown **mean**, unknown **variance**

$$\tau \sim \text{Gamma}(\Phi, \Psi)$$

$$\mu|\tau \sim \text{Normal}(M, \eta/\tau)$$

$$w|\mu, \tau \sim \text{Normal}(\mu, 1/\tau)$$

The posterior is

$$\tau|w \sim \text{Gamma}(\Phi', \Psi')$$

$$\mu|w, \tau \sim \text{Normal}(M', \eta'/\tau)$$

$$\Phi' = \dots\dots$$

$$\Psi' = \dots\dots$$

$$M' = \dots\dots$$

$$\eta' = \dots\dots$$